

Goal-Scoring Efficiency of Utah HC Players

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1 Introduction

In the fast-paced and highly competitive world of hockey, the ability to convert the shot opportunities into goals is a critical factor that often determines the outcome of games. Understanding which players and teams excel in this area provides valuable insights for coaches, analysts, and team managers, enabling them to make more informed decisions.

This analysis focuses on the development of an Expected Goals (xG) model and utilizes the xG metric to evaluate the goal-scoring efficiency of players on the current Utah HC roster.

2 Methodology

2.1 Data Collection

The analysis started with the collection of game data from the last four NHL seasons—2020/2021, 2021/2022, 2022/2023, and 2023/2024—focusing on player shot attempts and goals. This data was sourced from the official NHL API, which provides detailed play-by-play information, including the shooter’s identifier, shot location, and shot type. Before the analysis, certain shots were filtered out: blocked shots, shots from a team’s own half of the rink, shots without a recorded shot type, and shots from shootouts. As a result, the final dataset consisted of over 435,000 rows, capturing shots from 5,513 games, all 32 teams, and 1,322 shooting players.

2.2 Explanatory Data Analysis

Before developing the xG model, an Exploratory Data Analysis (EDA) was conducted to gain a deeper understanding of the data and inform subsequent modeling decisions. The EDA process included the following steps:

- **Data Cleaning:** The process included handling missing values, standardizing shot coordinates, and performing feature engineering. Features engineered included distance from goal, angle between the shooting player and the goal, time between shots, rebound identification, zone type, and strength type.
- **Visualization:** Histograms, bar plots, and heatmaps, were used to explore the distribution of shots, the frequency of different shot types, and the spatial distribution of shots on the ice. These visuals provided insights into common shooting patterns and areas of high goal-scoring probability.
- **Outlier Detection:** Players with very few shots were considered outliers and their shots were excluded from the analysis. A 5% quantile threshold was used for this purpose.

2.3 Expected Goals (xG) Model

A central component of this analysis was the creation of the custom xG model. During the development process, the following four models were created.

1. Baseline Probability Model: During the training phase, the goal probability for each combination of X and Y coordinates is calculated, but only for combinations with at least 10 shots. In the inference phase, the model returns the goal probability for a given combination of X and Y coordinates. If the combination is not found, the model returns a probability of zero.
2. Coordinates and Strength Probability Model: This model is similar to Model 1, with the key difference being that it calculates the goal probability based on a combination of X and Y coordinates as well as the strength type. The model requires at least 3 shots for a combination to return a non-zero probability.
3. Model 1 and 2 Combination: This model returns predictions from Model 2 when the combination of X and Y coordinates and strength type is known. If the combination is not available, it defaults to the prediction from Model 1.
4. Machine Learning-Based Model: This model is built using the following features: X and Y coordinates, distance from goal, angle between shooter and goal, period, time between shots, empty net flag, rebound flag, shot type, strength type, and zone type. The model employs a pipeline that begins with preprocessing features by categorizing them into categorical and numerical types. After preprocessing, the features are combined and used to train a decision tree classifier.

The models were compared, and Model 4 was selected as the best based on the following metrics: Logarithmic Loss, Brier Score, and ROC AUC Score.

	Log Loss	Brier Score	ROC AUC Score
Model			
1	0.256	0.063	0.738
2	0.342	0.062	0.751
3	0.235	0.062	0.760
4	0.181	0.048	0.869

Figure 1: Models comparison.

The model predictions were assigned as xG values to each shot attempt, representing the probability of scoring. Higher xG values were assigned to high-quality shots, while lower xG values indicated more challenging scoring opportunities.

3 Analysis and Findings

3.1 Comparison of Actual Goals to Expected Goals

With the xG model in place, the next step was to evaluate player efficiency by comparing each player’s total actual goals scored with their total xG.

The first comparison was made using the Goals Over Expected (GOE) metric, which is calculated by subtracting a player’s xG from their actual goals scored. A positive GOE indicates that a player is scoring more goals than expected, while a negative GOE suggests underperformance relative to their shot quality.

As shown in Figure 2, two players on the current roster, Clayton Keller and Michael Carcone, stood out during the 2023/2024 season, each scoring 8 goals more than the model predicted. Their GOE values are significantly higher than the rest of the team, although still not the highest in the league. The top player in the NHL by this metric for the 2023/2024 season was Artemi Panarin of the NY Rangers, with a GOE of 14.4.



Figure 2: Goals Over Expected, Utah HC, 2023/2024 season

However, since these players scored different numbers of goals from varying numbers of shots—each contributing to their respective xG values—these GOE numbers are difficult to compare directly. To address this, we can use the xG Conversion Rate, which normalizes the values and makes them more comparable. This metric, calculated as (Actual Goals / xG), represents the share of a player’s xG that has been converted into actual goals, providing insight into a player’s ability to capitalize on scoring opportunities.

In Figure 3, we can see that the ranking of the top 3 players changes significantly when using the xG Conversion Rate instead of the GOE. The most efficient shooter is now defenseman Michael Kesselring. Although he scored only 5 goals in the 2023/2024 season, his shots were assigned very low xG values because most were taken from behind the blue line, as shown in Figure 4. These are shots that the model considers more challenging scoring opportunities.

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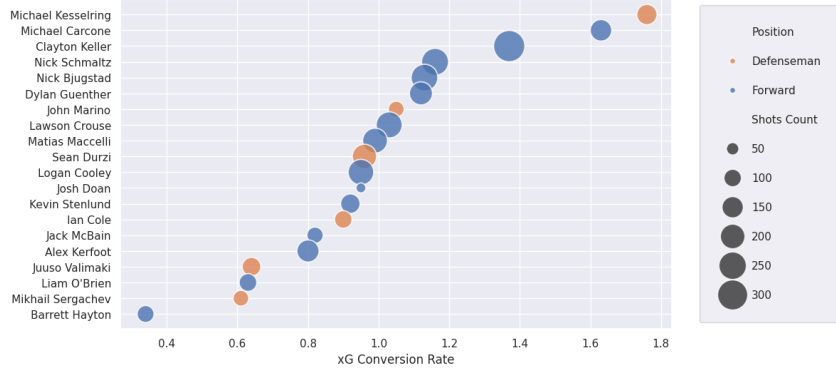


Figure 3: xG Conversion Rate per Player, Utah HC, 2023/2024 season

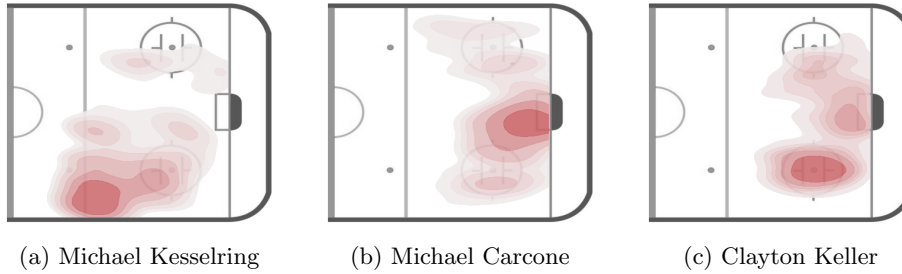


Figure 4: Shots heatmap, 2023/2024 season

When we look at the forwards, the order also changes when using the xG Conversion Rate. The most efficient-shooting forward is now Michael Carcone, with an xG Conversion Rate of 1.63, meaning he scored 63% more goals than expected in the 2023/2024 season. Clayton Keller finished the season with an xG Conversion Rate of 1.37. The Canadian also excelled in shooting percentage, converting 14% of his shots into goals, making him the second-best on the team in this statistic, right behind Josh Doan, who scored on 16% of his shots.

Given that Michael Carcone played nearly half the time and took half the shots compared to Clayton Keller over the entire season, it suggests that with more ice time or encouragement to shoot more, he could potentially become the team's most productive striker.

3.2 Expected Goals Over the Season

However, these values are aggregated over the entire season. Does this mean that the most efficient forwards consistently exceeded expectations throughout the 2023/2024 season? Figure 5 suggests otherwise.

Although Michael Carcone had an impressive start of the season, scoring 14 goals and achieving a GOE metric just below 10 by mid-December 2023, his

performance declined significantly, with only one goal in the following 10 weeks. While he regained his scoring form at the beginning of March 2024, he was unable to maintain it and finished the season with a total of 21 goals.

Clayton Keller, on the other hand, had a different trajectory. He began the season slowly, scoring roughly as expected, but demonstrated his significant talent by scoring 8 goals between Christmas and the All-Star Weekend. This period was crucial in boosting his GOE value. After returning from Toronto, Clayton Keller's performance aligned with expectations, and he finished the season with a GOE value similar to what he had at the end of January 2024.

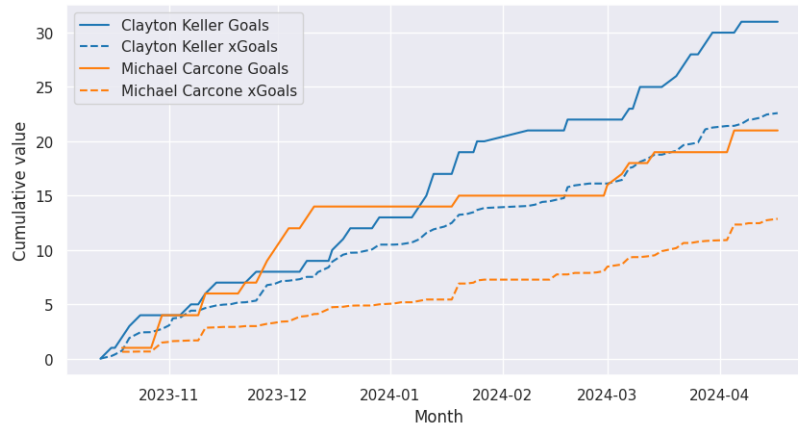


Figure 5: Cumulative Goals vs. Expected Goals, Clayton Keller and Michael Carcone, 2023/2024 season

4 Conclusion

This analysis evaluated the goal-scoring efficiency of players on the Utah HC roster using a custom xG model.

The Goals Over Expected (GOE) metric highlighted Clayton Keller and Michael Carcone as standout performers for the 2023/2024 season, with both players exceeding their xG predictions. This indicates their ability to capitalize on scoring chances beyond what the xG model anticipated.

Despite their superior efficiency in converting chances, their performance trends varied throughout the season. Michael Carcone experienced early-season success followed by a significant decline, while Clayton Keller showed steady improvement. This underscores the importance of monitoring performance across different periods and dimensions, rather than relying solely on aggregated seasonal statistics, which may not provide a complete picture.

Future analyses could explore how additional factors, such as team dynamics, assists, and opponent strength, impact goal-scoring efficiency.